TSA 2023

Problem Solving

Questions taken from the TSA Oxford 2023 Section 1 question paper

Questions	Total marks	Time limit
25	25	45 minutes

Instructions

- You have **45 minutes** to answer all **25 questions**.
- Each question is worth **one mark**. There are no penalties for incorrect answers, so attempt every question.
- Each question has five options, A to E. Choose the one option you consider correct.
- Calculators are **not** permitted. You may use this paper for rough working.
- Questions are numbered 1 to 25 for this set. The number in brackets shows the question's number in the original TSA 2023 paper, e.g. **1 (Q3)**.

How the time limit was set

The time limit follows the TARA Problem Solving module (22 questions in 40 minutes), scaled to the 25 questions in this set: 25 × 40 ÷ 22 = 45.45 minutes, rounded to the nearest minute (45 minutes).

Score: _____ / 25

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1 (Q2) The table shows the number of cold drinks sold, per hour, yesterday at a 24-hour service station.

	cold drinks
day time (15 hours)	18
night time (9 hours)	12

The service station also sells hot drinks. One-third fewer hot drinks were sold per hour than cold drinks, regardless of the time of day.

How many drinks in total were sold yesterday?

A 252

B 378

C 450

D 630

E 720

2 (Q6) The 4-digit PIN of my bank card has been set so that the first two digits and the last two digits both separately add up to 11.

The first digit is odd and the last digit in the PIN is the smallest of the four. All four digits are different, and no two digits in the PIN are numerically consecutive with each other. So, for example, if the PIN contains the digit ‘6’ then it cannot contain either the digits ‘5’ or ‘7’.

What is the first digit of my PIN?

A 1

B 3

C 5

D 7

E 9

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3 (Q7) Jarold weighs 60 kg. He decides to start a daily exercise regime of fast walking for 10 minutes, slow cycling for 20 minutes and slow swimming for 20 minutes.

The table below shows the rate at which calories are used up, per 30 minutes of exercise, by people of different weights.

	50 kg	*55 kg*	*60 kg*	*65 kg*
slow cycling	160	170	180	190
fast cycling	190	210	230	240
slow swimming	190	200	210	220
fast swimming	220	250	280	295
slow walking	50	55	60	75
fast walking	100	110	120	150

How many calories will Jarold use during exercise on the first day of this regime?

A 170 calories

B 190 calories

C 300 calories

D 340 calories

E 510 calories

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4 (Q8) As I waited for a train yesterday I was puzzled when I noticed the digital clock on the platform change from



to



but then a minute later it changed to



And then, after another minute, it changed to



I realised that the last digit was not displaying properly.

What was wrong with the last digit?

A One element was permanently on and one element was permanently off.

B One element was permanently on and two elements were permanently off.

C Two elements were permanently on and one element was permanently off.

D Two elements were permanently on and two elements were permanently off.

E Three elements were permanently on and one element was permanently off.

5 (Q12) David set out at 09:00 on a 10 km path up a hill, walking at 3 km/hour. Andrew set out from the top of the hill one hour later, walking the same path in the opposite direction, at 4 km/hour.

How far apart will David and Andrew be 1 hour after they pass each other?

A 1 km

B 3 km

C 4 km

D 6 km

E 7 km

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6 (Q13) Juan needs to buy some boxes to store his collection of badges. The types of box that are available from his local shop are shown in the table below.

box material	*length* (cm)	*width* (cm)	*depth* (cm)	*capacity* (cm^3)	*price* ($)
plastic	30	8	2	450	12
plastic	20	6	4	450	13
wood	20	10	3	500	15
wood	30	5	4	550	18
wood	20	10	4	600	22

Juan wants two different boxes with a combined capacity of at least 1000 cm^3. He also wants to make sure that neither box has any side longer than 25 cm.

What is the least that Juan can pay for the boxes?

A $28

B $30

C $31

D $34

E $35

7 (Q14) Dan wants to buy 12 kg of rock salt to help keep his driveway clear of ice during the winter. He has looked online and found four different offers on the recommended retail price of a 1 kg bag.

1 Store 1 has 30% off the price of a 1 kg bag.

2 Store 2 has buy one and get one at half price.

3 Store 3 has buy two and get one free.

4 Store 4 has one third extra for same price as a 1 kg bag.

Which pair of offers works out to be the same price per kg?

A 1 and 2

B 1 and 4

C 2 and 3

D 2 and 4

E 3 and 4

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8 (Q18) Five friends play a word game called Wordsmith. The objective is to score as many as points as possible, using the letters of a randomly chosen starting word to create new words. Each letter used in a word scores one point, and the player with the most points after five rounds is the winner.

The word chosen for this particular game was ‘RAINBOW’. In the first four rounds, the words selected by the five players were as follows:

Georgina	*Riaz*	*Bobby*	*Vandy*	*Sonny*
BRAIN	BROWN	ROBIN	BRAWN	WORN
WARN	BARON	BORN	IRON	BOAR
BRAN	RAIN	WAR	BROW	BAN
NOW	WON	AN	NO	OR

The words given in the final round by the five players were: ‘ROW’, ‘BAR’, ‘AIR’, ‘ON’ and ‘I’ (not in order).

Which one of the following is **not** a possible difference between the final scores of the players with the most and least points?

A 1

B 2

C 3

D 4

E 5

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9 (Q19) I need to make sure that my three children all have a piece of fruit to take to school on each of the five days next week. Jessica and Bruce will not eat apples and Victoria will not eat oranges. My local shop sells a variety of bags containing either just apples, just oranges or different mixtures of the two fruits. The options available are shown below:

option	*apples*	*oranges*	*price (p)*	*total number in bag*
apple bag	6	0	99	6
orange bag	0	6	109	6
lucky dip	at least 1	at least 1	95	6
half and half	3	3	105	6

What is the minimum that I could spend to be **sure** that I have enough fruit for all three children next week?

A £2.85

B £2.99

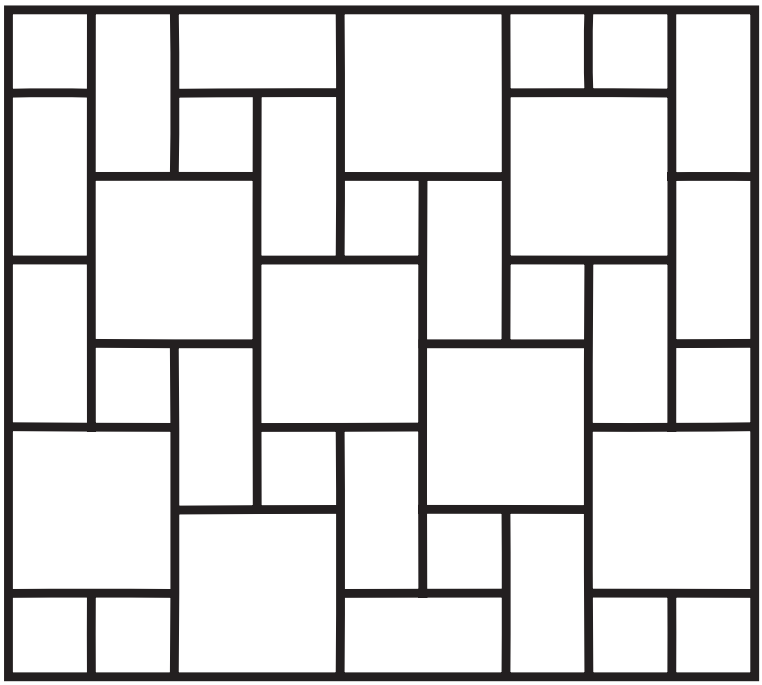
C £3.04

D £3.17

E £3.19

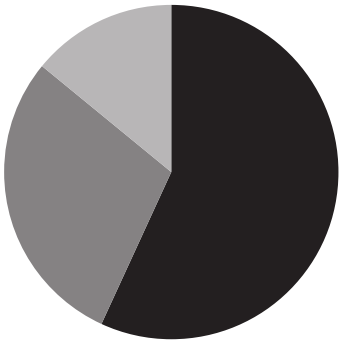
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10 (Q20) The block paved area shown below is laid using three different-sized blocks.

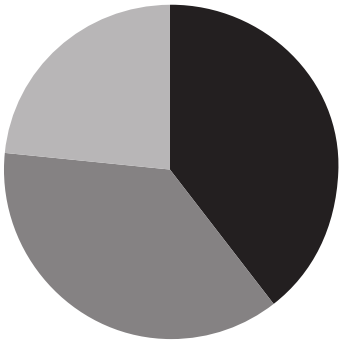


Which one of the following charts shows the relative comparison of the number of each size of block used in this paved area?

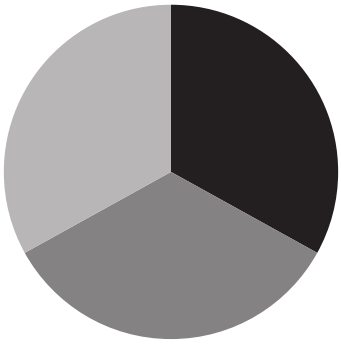
A



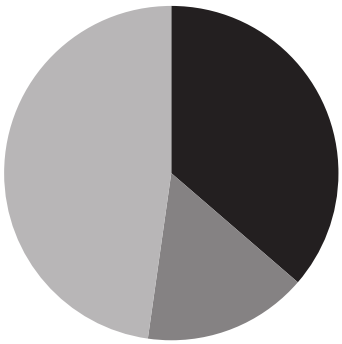
B



C



D



E



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11 (Q24) Every Christmas season, an ice-skating rink is laid in the central square in Boland. The rink is open to the public from 12th December to 9th January inclusive, between 2 pm and 5 pm. It is closed on Sundays and Christmas Day (25th December), but there is an extra session on Saturdays between 10 am and 1 pm. The charge is £2 an hour.

When Jimmy was learning to skate, he went to the rink for two hours every session that it was open. That year, 12th December was on a Monday.

What is the total amount that Jimmy had to pay?

A £100

B £104

C £108

D £112

E £116

12 (Q25) I need to create gift bags for the guests who are coming to my party this weekend. Each bag will contain 5 sweets, a small toy and a balloon. The prices for the items are shown in the table below.

	price per pack		
	first 10 packs	*next 20 packs*	*any further packs*
sweets (20 in a pack)	\$0.45	\$0.40	\$0.35
balloons (20 in a pack)	\$0.90	\$0.80	\$0.70
toys (1 in a pack)	\$3.50	\$3.00	\$2.50

I need 23 gift bags for my party and do not wish to buy any more of any item than is necessary.

What will be the total price I will pay for the items in my gift bags?

A \$73.00

B \$78.50

C \$85.50

D \$95.10

E \$103.10

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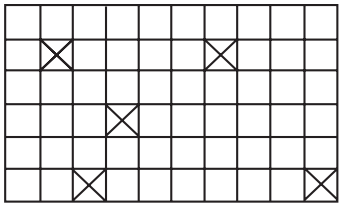
13 (Q26) A roofer is looking at the front-facing roofs of four houses. The survey finds five broken tiles on the roof of each house. The positions of the broken tiles are shown in the table below, with the column (col) and row numbers starting from the bottom left-hand tile on each roof.

	broken tile 1		*broken tile 2*		*broken tile 3*		*broken tile 4*		*broken tile 5*	
	col	*row*	*col*	*row*	*col*	*row*	*col*	*row*	*col*	*row*
house W	2	2	4	5	6	2	8	5	10	2
house X	2	3	5	4	6	3	8	5	9	2
house Y	2	5	3	1	4	3	7	5	10	1
house Z	1	2	3	5	4	2	6	4	8	5

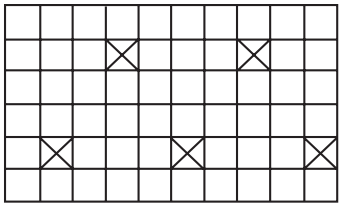
The roofer also draws a diagram for each of the roofs, showing the position of each of the broken tiles marked with an X, with the bottom left-hand tile of the roof in the bottom left-hand corner of the diagram.

Which one of the diagrams shown below does **not** correspond to one of the four houses?

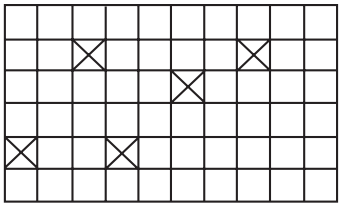
A



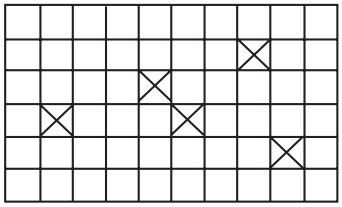
B



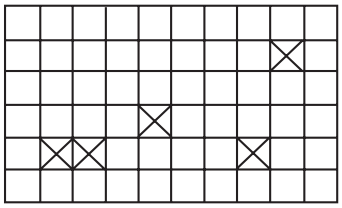
C



D



E



14 (Q30) Tabatha has a block of cheese with the dimensions 12 x 10 x 5 cm. She cuts it into 1 cm cubes for a party. She forgets to cut the rind (tough outside layer) off before she cuts the block into cubes. As she doesn't have a lot of time before her guests start to arrive she only removes the rind from cheese cubes which have rind on two or more faces.

From how many cheese cubes will she remove rind?

A 84

B 92

C 100

D 108

E 192

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15 (Q31) Imani is looking to book a four-night stay in May at the Highgate Hotel, to include a Sunday night so that she can attend an evening orchestral concert.

The table below shows the daily rates for the Highgate Hotel.

Highgate Hotel
Daily rates for a single room with breakfast
May

Monday	*Tuesday*	*Wednesday*	*Thursday*	*Friday*	*Saturday*	*Sunday*
1st **£30**	2nd **£40**	3rd **£50**	4th **£50**	5th **£20**	6th **£50**	7th **£40**
8th **£30**	9th **£40**	10th **£50**	11th **£50**	12th **£20**	13th **£50**	14th **£40**
15th **£30**	16th **£40**	17th **£60**	18th **£40**	19th **£30**	20th **£40**	21st **£40**
22nd **£30**	23rd **£20**	24th **£50**	25th **£40**	26th **£30**	27th **£60**	28th **£30**
29th **£40**	30th **£50**	31st **£50**				

May Special Offers
Book Saturday and Sunday and stay Friday for free.
Book Sunday and Monday and stay Tuesday for free.
Only one special offer can be used per booking.

What is the least amount Imani needs to pay for her four-night stay?

A £100

B £110

C £120

D £130

E £140

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16 (Q32) In a particular coding system, letters of the alphabet are represented by two-character or three-character combinations of the symbols o, + and –.

For example, the letter A is represented as o+, and the letter Z is represented as –o+.

The table shows how certain words are represented in this coding system.

word	*code*
AND	o+–++–o
MAN	–+–o+–++
MEND	–+–++–++–o
RUNNY	oo+o–++–++ooo

How is the word EAR represented by the code?

A ++o+oo

B –++o+oo

C ++o+–oo

D ++o+oo+

E –++o+–oo+

17 (Q36) Becky works from home making soft toys which she sells online. Each one takes her exactly 16 minutes to make and she takes a 5-minute break between finishing one item and starting the next. She always takes a 40-minute break for her lunch. Today she is starting on a new batch of toys and is planning to work from 08:30 until 17:30.

How many soft toys will she have completed by 17:30?

A 23

B 24

C 25

D 26

E 27

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18 (Q37) Last Saturday a large number of men and women competed in a long-distance cycling event. The time that each competitor took to complete the course was recorded. The following table shows the range of times taken and the percentages of men and women who achieved these times.

time in minutes	*percentage of men*	*percentage of women*
0–120	5	1
121–150	5	5
151–180	6	4
181–210	12	21
211–240	30	15
241–300	26	36
301–420	16	12
421 or more	0	6

There were twice as many men as women in the event.

What percentage of all the competitors took more than four hours to complete the event?

A 46%

B 48%

C 54%

D 69%

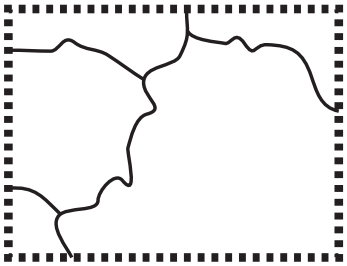
E 71%

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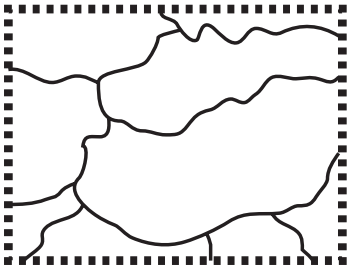
19 (Q38) The following pictures are blank maps. The continuous lines indicate country borders, the dotted lines show the borders of the maps. These maps need to be painted so that each country is filled with one colour and that no two neighbouring countries (countries with a common border) have the same colour. The fewest number of colours should be used on a given map.

Which one of the following blank maps will require 4 different colours to be painted?

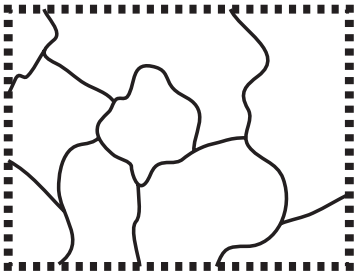
A



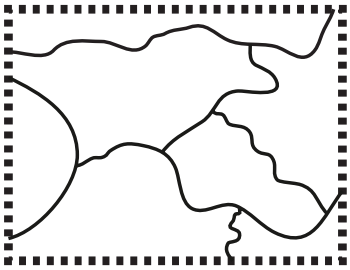
B



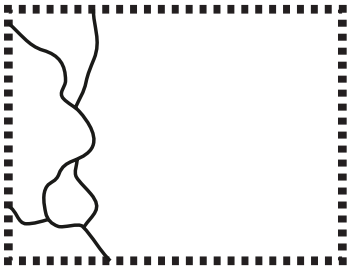
C



D



E



20 (Q42) Isaac has been on a diet to lose weight. He wants to be certain how much weight he has lost, so he checks himself on three different weighing scales consecutively. One reads 95 kg, one reads 93 kg, and one reads 96 kg. Each weighing scale is accurate to within 2 kg above or below the value they actually read.

What is the minimum range in which Isaac's true weight must lie?

A 91–94 kg

B 91–98 kg

C 93–96 kg

D 94–95 kg

E 95–98 kg

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21 (Q43) A box contains equal numbers of each colour of red socks, white socks and blue socks. Additional red socks are put into the box which doubles the number of red socks in the box. 24 of the white socks are removed from the box and replaced with 24 blue socks.

If there are now four times as many blue socks as white socks in the box, how many red socks are there in the box?

A 40

B 60

C 64

D 72

E 80

22 (Q44) Three numbers are combined in a particular way, using addition, subtraction, multiplication and/or division.

The numbers 3, 4 and 18 (used once each) are combined to give the number 10.

Which one of the following sets of numbers combines the first three numbers, in the same way to produce the final number?

A 2, 5 and 12 to give 11

B 4, 6 and 12 to give 8

C 5, 12 and 20 to give 18

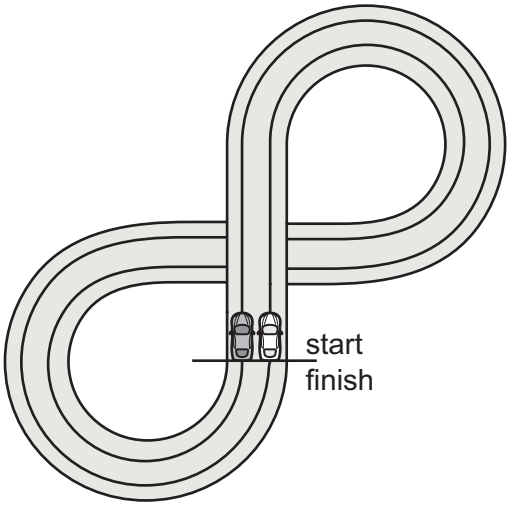
D 6, 8 and 12 to give 14

E 8, 16 and 24 to give 32

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23 (Q46) The drawing shows a slot car track.

A grey car and a white car start together at the same point and travel in the same direction. Because of the figure-of-eight arrangement of the track, each car travels exactly the same distance to complete one lap of the track. The grey car is 12% faster than the white car.



How many laps will the grey car have completed before it passes the white car for the first time?

A 8

B 9

C 10

D 11

E 12

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24 (Q48) The charges at the Puffins adventure park are shown in the following table.

	price ($)
adult entry	30
child entry	25
family ticket (entry for 2 adults and 2 children, includes 8 single-ride tokens)	120
single-ride token	5
super ride (10 single-ride tokens)	40

A single-ride token allows one person one ride on any of the adventure rides.

Mike and Helen are visiting Puffins with their two children, Amy and Ben, and Helen's mother, Pearl. They pay $250 and want to have as many rides as possible. Mike, Helen, Amy and Ben will each have the same number of rides. Pearl will have at least two rides.

What is the maximum number of rides that Amy will have?

A 4

B 5

C 6

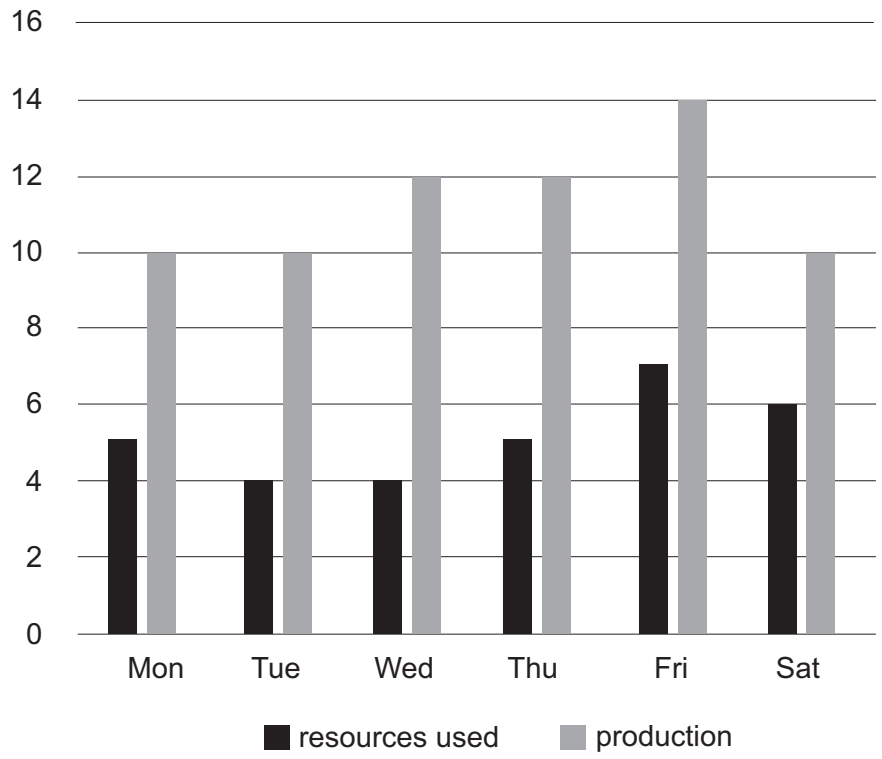
D 7

E 8

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25 (Q50) At a factory, the senior management are monitoring the ratio of production to resources used each day. For example, if on one day the production is $200 000 and the resources used is $40 000, then the ratio is 5.

Last week, the production and resources used each day (both in tens of thousands of dollars) were as shown in the graph below.



Between which two days did the ratio of production to resources used increase by the greatest percentage?

A Monday to Tuesday

B Tuesday to Wednesday

C Wednesday to Thursday

D Thursday to Friday

E Friday to Saturday

END OF TEST

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